Transfer: gravity and motion

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Apply gravity, projectile and orbital reasoning to a new set of cases.

SUCCESS CRITERIA - I can...

1. Choose a diagram or equation. State what it treats as true before calculating.
2. Connect diagrams to forces and motion.
3. Check units and explain the limits of conclusions.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “model” and “gravity” to explain your answer. Sentence starters are optional.

model: a drawing, explanation or equation for a physical situation

gravity: an attractive interaction between masses

projectile: an object moving after launch under the stated forces

01 Warm-up: bring back an earlier idea

A circular orbital radius becomes **9** times as large around the same source. **Compare orbital speed** and period.

Word help: radius = distance from the center to the edge of a circle or sphere

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $F_g = mg$; $a_c = v^2/r$; $h = \frac{1}{2}gt^2$; range = $v_x\,t$.
Match each equation to its case: near-surface weight, circular motion, or horizontal launch. Use g = **10 m/s²**.

Worked reasoning: A toy horizontal projectile falls for **2 s** while moving at **4 m/s**. Range is **8 m**. The vertical motion sets time; the horizontal motion sets range once time is known.

Your turn: Draw two coordinated timelines, one for each direction, with a shared **2 s** endpoint.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .



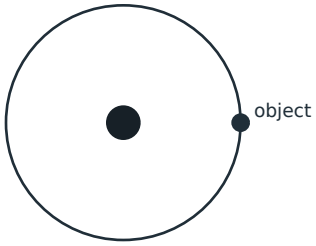
Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

A satellite moves in a circular orbit.
1. **Draw velocity** and gravity arrows.
2. **Explain why floating** does not mean zero gravity.

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .



The center is the attracting body; the small dot on the circle is the orbiting object.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

A ball launches horizontally at **6 m/s** from **20 m**. **Find time**, range, final vertical velocity and final speed. **Label the diagram** with the known quantities before calculating. Ignore air resistance; use g = **10 m/s²** when needed.

Sentence starter: I found … [unit]. This value means … in this situation.



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

05 Explain your reasoning

A Sun-orbit returns every **2 years**. Why can period alone not prove an Earth encounter? **Name one measurement or observation** you would request.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

For fixed masses, what separation multiplier makes gravity **1/16** of its original value? What multiplier is needed if both masses also become **6** times as large? **Label the masses**, field or force relevant to the question.

Word help: separation = distance between the specified objects or centers

Sentence starter: In my work, I used … to show … . I labeled it … .



Label separation if needed.

Source Object

Marzano check - circle: ① ② ③

07 Exit check

For GM=**400 m³/s²** and circular r=**25 m**, find speed, acceleration and period. **Explain what changes** in velocity. **Label the diagram** with the known quantities before calculating. Ignore air resistance; use g = **10 m/s²** when needed.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Separate horizontal and vertical motion

Label x and y; state positive directions.

Velocity:

Acceleration / force:

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

A horizontal projectile launches from a **45 m** platform at **12 m/s**; g=**10 m/s²**. **Find flight time**, range and final velocity. Then halve g with all starting values fixed and **compare range**. **Label the diagram** with the known quantities before calculating. Ignore air resistance; use g = **10 m/s²** when needed.

Sentence starter: Both … . They differ in … because … .



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③